

Lecture 11

Orthogonality General Inner Products

Consider $\mathbb{R}^4$ and a subspace $\mathcal{U}$ with the following two bases.

check
 $\langle \underline{u}_1, \underline{u}_2 \rangle = 4 \neq 0$
 $\therefore B$ is not orthogonal

$$B = \left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \right] = [\underline{u}_1, \underline{u}_2, \underline{u}_3]$$

$$Q = \left[\begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} \right] = [\underline{q}_1, \underline{q}_2, \underline{q}_3]$$

Let $\underline{a} = (1, 6, 1, 8)^T$. What is the formula for $\text{Proj}_{\mathcal{U}}(\underline{a})$?

1. $\text{Proj}_{\mathcal{U}}(\underline{a}) = \frac{\langle \underline{a}, \underline{u}_1 \rangle}{\langle \underline{u}_1, \underline{u}_1 \rangle} \underline{u}_1 + \frac{\langle \underline{a}, \underline{u}_2 \rangle}{\langle \underline{u}_2, \underline{u}_2 \rangle} \underline{u}_2 + \frac{\langle \underline{a}, \underline{u}_3 \rangle}{\langle \underline{u}_3, \underline{u}_3 \rangle} \underline{u}_3 = \begin{bmatrix} 2 \\ 5 \\ 2 \\ 5 \end{bmatrix}$

✓ 2. $\text{Proj}_{\mathcal{U}}(\underline{a}) = \frac{\langle \underline{a}, \underline{q}_1 \rangle}{\langle \underline{q}_1, \underline{q}_1 \rangle} \underline{q}_1 + \frac{\langle \underline{a}, \underline{q}_2 \rangle}{\langle \underline{q}_2, \underline{q}_2 \rangle} \underline{q}_2 + \frac{\langle \underline{a}, \underline{q}_3 \rangle}{\langle \underline{q}_3, \underline{q}_3 \rangle} \underline{q}_3 = \begin{bmatrix} 1 \\ 7 \\ 1 \\ 7 \end{bmatrix}$

Recall: Projection formula works if basis is orthogonal i.e. check $\langle \underline{v}_i, \underline{v}_j \rangle = 0$ if $i \neq j$

check $\left. \begin{aligned} \langle \underline{q}_1, \underline{q}_2 \rangle &= 0 \\ \langle \underline{q}_1, \underline{q}_3 \rangle &= 0 \\ \langle \underline{q}_2, \underline{q}_3 \rangle &= 0 \end{aligned} \right\}$ must all satisfy

$\therefore Q$ is orthogonal

$\therefore$ Basis does MATTER!

Lecture 11

Orthogonality

- Orthogonal Basis
 - >> Extract Coefficients, Orthogonal Projection
- Gram-Schmidt Procedure
 - >> Orthogonal Sets/Basis, QR Decomposition

General Inner Product (Can functions be orthogonal?)

↳ Extend inner product to abstract vectors.

Theorem. Let $\mathcal{V}$ be an inner product space and consider a subspace $\mathcal{U}$.

Let $\mathcal{A} = [\mathbf{v}_1, \dots, \mathbf{v}_k]$ be an orthogonal basis for $\mathcal{U}$. Then

$$\text{Proj}_{\mathcal{U}}(\mathbf{a}) = \frac{\langle \mathbf{a}, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 + \frac{\langle \mathbf{a}, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle} \mathbf{v}_2 + \dots + \frac{\langle \mathbf{a}, \mathbf{v}_k \rangle}{\langle \mathbf{v}_k, \mathbf{v}_k \rangle} \mathbf{v}_k \text{ for any } \mathbf{a} \in \mathcal{V}.$$

Example.

Consider $\mathbb{R}^4$ and a subspace $\mathcal{U}$ described by the basis: $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \right] = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3]$ *usually not given orthogonal*

Let $\mathbf{a} = (1, 6, 1, 8)^T$. What is $\text{Proj}_{\mathcal{U}}(\mathbf{a})$?

⚠ $\mathcal{B}$ is NOT orthogonal, then Proj. formula fails

Task: Given any basis $\mathcal{B} = [\mathbf{u}_1, \dots, \mathbf{u}_k]$

WANT: orthogonal basis $\mathcal{A} = [\mathbf{v}_1, \dots, \mathbf{v}_k]$ s.t. $\text{span}(\mathcal{A}) = \text{span}(\mathcal{B})$

Definition.

A set $\mathcal{A} = \{v_1, \dots, v_k\}$ is orthogonal if $\langle v_i, v_j \rangle = 0$ for $i \neq j$.

Proposition. Let $\mathcal{V}$ be a vector space which has an inner product. An orthogonal set of nonzero vectors is also linearly independent.

$\mathcal{A}$ orthogonal $\Rightarrow \mathcal{A}$ is linearly independent

Proof $\lambda_1 v_1 + \dots + \lambda_k v_k = 0$ ———— (*)

each of v_i is
non-zero vector

$\langle v_i, v_i \rangle \neq 0$
(because of
definiteness)

Take inner product with v_i ,

$$\langle \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k, v_i \rangle = \langle 0, v_i \rangle = 0$$

$$\lambda_1 \underbrace{\langle v_1, v_i \rangle}_{=0} + \lambda_2 \underbrace{\langle v_2, v_i \rangle}_{=0} + \dots + \lambda_k \underbrace{\langle v_k, v_i \rangle}_{=0} = 0 \Rightarrow \lambda_i \langle v_i, v_i \rangle = 0 \Rightarrow \lambda_i = 0$$

$\Rightarrow$ set is orthogonal

$\lambda_i = 0$ for all i

Example. $\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\}$ is orthogonal. So, $\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$.

Therefore, we can write any tuple in $\mathbb{R}^2$ as a unique linear combination of $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$.

Find the linear combination for $\begin{bmatrix} 50 \\ 75 \end{bmatrix}$.
Aka:
- extract coefficients
- decomposition

$$\lambda_1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \lambda_2 \begin{bmatrix} -4 \\ 3 \end{bmatrix} = \begin{bmatrix} 50 \\ 75 \end{bmatrix} \quad \text{- finding coordinate vector.}$$

Take inner product with $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$$\text{LHS} = \left\langle \lambda_1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \lambda_2 \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\rangle = \lambda_1 \left\langle \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\rangle + \lambda_2 \left\langle \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\rangle = 25\lambda_1$$

$$\text{RHS} = \left\langle \begin{bmatrix} 50 \\ 75 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\rangle = 450$$

$$25\lambda_1 = 450 \Rightarrow \lambda_1 = 18$$

Similarly, take inner prod with $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$... find λ_2 .

Definition.

An orthogonal set $\mathcal{A} = \{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is *orthonormal* if $\|\mathbf{q}_i\| = 1$ for all i .

Example. $\left[\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right]$ is an orthogonal basis for $\mathbb{R}^2$.

Is $\left[\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right]$ an orthonormal basis?

$$\left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = 5 = \left\| \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\|$$

→ No!

An orthonormal basis is $\left[\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} \right]$.

Generally,
 $[\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_k]$ orthogonal
↓

$\left[\frac{\tilde{\mathbf{v}}_1}{\|\tilde{\mathbf{v}}_1\|}, \dots, \frac{\tilde{\mathbf{v}}_k}{\|\tilde{\mathbf{v}}_k\|} \right]$ orthonormal

Proposition. If $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is orthonormal and

$$\mathbf{v} = \lambda_1 \mathbf{q}_1 + \dots + \lambda_k \mathbf{q}_k,$$

then

$$\lambda_i = \langle \mathbf{v}, \mathbf{q}_i \rangle \text{ for all } i.$$

Corollary. If $\mathcal{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_k]$ is an orthonormal basis for vector space $\mathcal{V}$, then the coordinate vector of $\mathbf{v}$ with respect to $\mathcal{Q}$ is

$$[\mathbf{v}]_{\mathcal{Q}} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{q}_1 \rangle \\ \langle \mathbf{v}, \mathbf{q}_2 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{q}_k \rangle \end{bmatrix}.$$

Proof in coroll.

Gram-Schmidt Procedure

Input: Any basis $\mathcal{B} = [u_1, \dots, u_k]$ for $\mathcal{U}$.

Intermediate
step . orthogonal
basis



$[v_1, \dots, v_k]$ for $\mathcal{U}$

} Divide by $\|v_i\|$

Output: An orthonormal basis $\mathcal{A} = [q_1, \dots, q_k]$ for $\mathcal{U}$.

Gram-Schmidt Procedure

INPUT : $\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$

Find $\underline{v}_1, \underline{v}_2 \dots \underline{v}_k$

- Set $\mathbf{v}_1 = \mathbf{u}_1$

- For $j = 2, \dots, k$,

$$\mathbf{v}_j = \mathbf{u}_j - \frac{\langle \mathbf{u}_j, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 - \dots - \frac{\langle \mathbf{u}_j, \mathbf{v}_{j-1} \rangle}{\langle \mathbf{v}_{j-1}, \mathbf{v}_{j-1} \rangle} \mathbf{v}_{j-1}$$

- Normalise all $\mathbf{v}_i$. i.e. Set $\mathcal{A} = \left\{ \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \dots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right\}$

Example. Consider the Euclidean inner product on $\mathbb{R}^4$.

Apply Gram-Schmidt procedure on $\mathcal{B} = \left[\begin{array}{c} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \end{array} \right] !$

$\underline{u}_1 \quad \underline{u}_2 \quad \underline{u}_3$

**Order is very important for this procedure.*

What is $\underline{v}_1$?

Gram-Schmidt Procedure

INPUT : $\mathcal{B} = \{\underline{u}_1, \dots, \underline{u}_k\}$

- Set $\underline{v}_1 = \underline{u}_1$
- For $j = 2, \dots, k$,

$$\underline{v}_j = \underline{u}_j - \frac{\langle \underline{u}_j, \underline{v}_1 \rangle}{\langle \underline{v}_1, \underline{v}_1 \rangle} \underline{v}_1 - \dots - \frac{\langle \underline{u}_j, \underline{v}_{j-1} \rangle}{\langle \underline{v}_{j-1}, \underline{v}_{j-1} \rangle} \underline{v}_{j-1}$$
- Normalise all $\underline{v}_i$. i.e. Set $\mathcal{A} = \left\{ \frac{\underline{v}_1}{\|\underline{v}_1\|}, \dots, \frac{\underline{v}_k}{\|\underline{v}_k\|} \right\}$

$$\underline{v}_1 = \underline{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Example. Consider the Euclidean inner product on $\mathbb{R}^4$.

Apply Gram-Schmidt procedure on $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \right] !$

What is $\mathbf{v}_2$? $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \leftarrow$

HIDDEN

Gram-Schmidt Procedure

INPUT : $\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$

- Set $\mathbf{v}_1 = \mathbf{u}_1$
- For $j = 2, \dots, k$,

$$\mathbf{v}_j = \mathbf{u}_j - \frac{\langle \mathbf{u}_j, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 - \dots - \frac{\langle \mathbf{u}_j, \mathbf{v}_{j-1} \rangle}{\langle \mathbf{v}_{j-1}, \mathbf{v}_{j-1} \rangle} \mathbf{v}_{j-1}$$
- Normalise all $\mathbf{v}_i$. i.e. Set $\mathcal{A} = \left\{ \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \dots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right\}$

use $\mathbf{u}_2$ and $\mathbf{v}_1$ to find $\mathbf{v}_2$.

$$\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = 4 \quad \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 4$$

$$\mathbf{v}_2 = \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix} - \frac{4}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 3 \\ -3 \end{bmatrix}$$

Example. Consider the Euclidean inner product on $\mathbb{R}^4$.

Apply Gram-Schmidt procedure on $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \right] !$

What is $\mathbf{v}_3$? $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ -3 \\ 3 \\ -3 \end{bmatrix}$

HIDDEN

Gram-Schmidt Procedure

INPUT : $\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$

- Set $\mathbf{v}_1 = \mathbf{u}_1$
- For $j = 2, \dots, k$,

$$\mathbf{v}_j = \mathbf{u}_j - \frac{\langle \mathbf{u}_j, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 - \dots - \frac{\langle \mathbf{u}_j, \mathbf{v}_{j-1} \rangle}{\langle \mathbf{v}_{j-1}, \mathbf{v}_{j-1} \rangle} \mathbf{v}_{j-1}$$
- Normalise all $\mathbf{v}_i$. i.e. Set $\mathcal{A} = \left\{ \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \dots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right\}$

$$\langle \mathbf{u}_3, \mathbf{v}_1 \rangle = 4$$

$$\langle \mathbf{u}_3, \mathbf{v}_2 \rangle = 24$$

$$\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 4$$

$$\langle \mathbf{v}_2, \mathbf{v}_2 \rangle = 36$$

$$\mathbf{v}_3 = \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} - \frac{4}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{24}{36} \begin{bmatrix} 3 \\ -3 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 3 \\ -3 \end{bmatrix}$$

Example. Consider the Euclidean inner product on $\mathbb{R}^4$.

Apply Gram-Schmidt procedure on $\mathcal{B} = \left[\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix} \right] !$

What is an orthonormal basis?

HIDDEN

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad u_2 = \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix} \quad u_3 = \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad v_2 = \begin{bmatrix} 4 \\ -2 \\ 4 \\ -2 \end{bmatrix} \quad v_3 = \begin{bmatrix} 6 \\ 2 \\ 0 \\ -4 \end{bmatrix}$$

Gram-Schmidt Procedure

INPUT : $\mathcal{B} = \{u_1, \dots, u_k\}$

- Set $v_1 = u_1$

- For $j = 2, \dots, k,$

$$v_j = u_j - \frac{\langle u_j, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \dots - \frac{\langle u_j, v_{j-1} \rangle}{\langle v_{j-1}, v_{j-1} \rangle} v_{j-1}$$

- Normalise all v_i . i.e. Set $\mathcal{A} = \left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_k}{\|v_k\|} \right\}$

$$\|u_1\| = 2$$

$$\|u_2\| = 6$$

$$\|u_3\| = 6$$

$$\mathcal{A} = \left\{ \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ -1/2 \end{bmatrix} \right\}$$

HIDDEN

Gram-Schmidt Procedure

INPUT: $\mathcal{B} = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$

*Detailed proof
in video*

- Set $\mathbf{v}_1 = \mathbf{u}_1$

- For $j = 2, \dots, k$,

$$\mathbf{v}_j = \mathbf{u}_j - \frac{\langle \mathbf{u}_j, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 - \dots - \frac{\langle \mathbf{u}_j, \mathbf{v}_{j-1} \rangle}{\langle \mathbf{v}_{j-1}, \mathbf{v}_{j-1} \rangle} \mathbf{v}_{j-1}$$

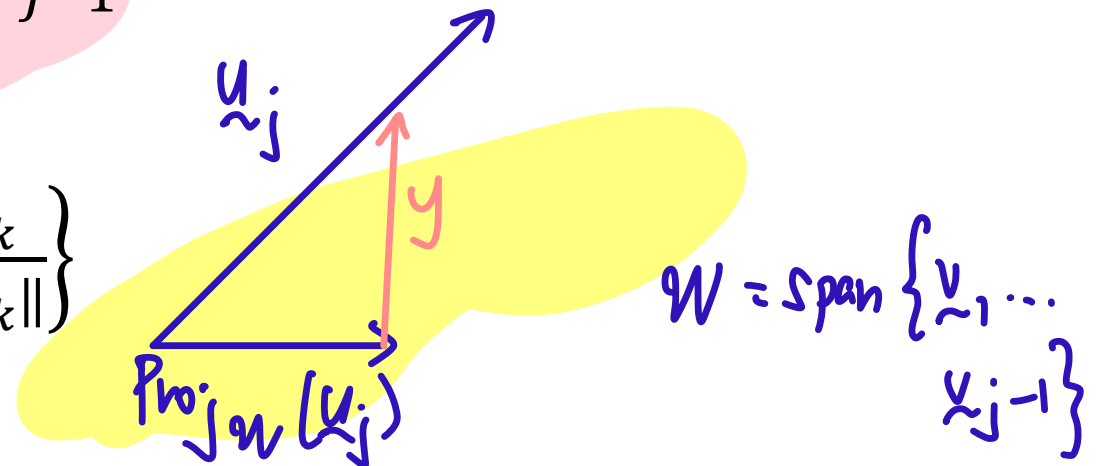
- Normalise all $\mathbf{v}_i$. i.e. Set $\mathcal{A} = \left\{ \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \dots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|} \right\}$

Why Does Gram-Schmidt Work?

Claims:

At step j ,

- $\mathbf{v}_j = \mathbf{u}_j - \text{Proj}_{\mathcal{W}}(\mathbf{u}_j)$ where $\mathcal{W} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{j-1}\}$.
- $\{\mathbf{v}_1, \dots, \mathbf{v}_j\}$ is orthogonal.
- $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\} = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_j\}$.



QR decomposition

$$\hat{Q} = \begin{bmatrix} | & | & \dots & | \\ \hat{q}_1 & \hat{q}_2 & \dots & \hat{q}_k \\ | & | & \dots & | \end{bmatrix}$$

Definition.

A matrix Q is *orthogonal* if its column vectors are orthonormal.

Proposition.

A matrix Q is orthogonal if and only if

$$Q^T Q = I.$$

$$\hat{Q}^T \hat{Q} = I$$

① If $\hat{q}_1, \dots, \hat{q}_k$ are "just" orthogonal,
the matrix has NO SPECIAL NAME.

Proof in class

Theorem. QR Decomposition.

Let U be ~~square~~ matrix whose columns are linearly independent.

Then there is an orthogonal matrix Q and upper triangular matrix R such that $U = QR$. Such a factoring is called the QR decomposition.

Definition.

A square matrix Q is *orthogonal* if its column vectors are orthonormal. Equivalently, we have

$$Q^T Q = I.$$

Example. Find the QR decomposition of the matrix.

$$U = \begin{bmatrix} 1 & 4 & 6 \\ 1 & -2 & 2 \\ 1 & 4 & 0 \\ 1 & -2 & -4 \end{bmatrix}.$$

→ use Gram Schmidt on $[u_1, u_2, u_3]$ to find $[q_1, q_2, q_3]$

$$Q = \begin{matrix} q_1 & q_2 & q_3 \\ \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \end{matrix}$$

want to find $\hat{R}$ s.t. $\hat{U} = Q\hat{R}$ → upper triangular

- Actually, entries of $\hat{R}$ can be obtained from Gram-Schmidt Procedure

Theorem. QR Decomposition.

Let U be square matrix whose columns are linearly independent. Then there is an orthogonal matrix Q and upper triangular matrix R such that $U = QR$. Such a factoring is called the QR decomposition.

Definition.

A matrix Q is *orthogonal* if its column vectors are orthonormal. Equivalently, $Q^T Q = I$.

Example. Find the QR decomposition of the matrix.

$$U = \begin{bmatrix} 1 & 4 & 6 \\ 1 & -2 & 2 \\ 1 & 4 & 0 \\ 1 & -2 & -4 \end{bmatrix}$$

Found during
Gram-Schmidt

$$\begin{array}{ccc} \tilde{u}_1 & \tilde{u}_2 & \tilde{u}_3 \\ \begin{bmatrix} 1 & 4 & 6 \\ 1 & -2 & 2 \\ 1 & 4 & 0 \\ 1 & -2 & -4 \end{bmatrix} & = & \begin{array}{ccc} \tilde{v}_1 & \tilde{v}_2 & \tilde{v}_3 \\ \begin{bmatrix} 1 & 3 & 3 \\ 1 & -3 & 3 \\ 1 & 3 & -3 \\ 1 & -3 & -3 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2/3 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}$$

upper triangular

$$\tilde{u}_1 = \tilde{v}_1 \rightarrow \text{from formula}$$

$$\tilde{u}_2 = \tilde{v}_2 + \tilde{v}_1$$

$$\tilde{u}_3 = \tilde{v}_3 + \frac{2}{3}\tilde{v}_2 + \tilde{v}_1$$

Theorem. QR Decomposition.

Let $\mathbf{U}$ be ~~square~~ matrix whose columns are linearly independent. Then there is an orthogonal matrix $\mathbf{Q}$ and upper triangular matrix $\mathbf{R}$ such that $\mathbf{U} = \mathbf{QR}$. Such a factoring is called the QR decomposition.

Definition.

A matrix $\mathbf{Q}$ is *orthogonal* if its column vectors are orthonormal.

$$\begin{bmatrix} 1 & 4 & 6 \\ 1 & -2 & 2 \\ 1 & 4 & 0 \\ 1 & -2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & -3 & 3 \\ 1 & 3 & -3 \\ 1 & -3 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2/3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \\ 1/2 & -1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2/3 \\ 0 & 0 & 1 \end{bmatrix}$$

Example. Find the QR decomposition of the matrix.

$$\mathbf{U} = \begin{bmatrix} 1 & 4 & 6 & 4 \\ 1 & -2 & 2 & 0 \\ 1 & 4 & 0 & 0 \\ 1 & -2 & -4 & 0 \end{bmatrix}.$$

$$= \mathbf{Q} \begin{bmatrix} 2 & 2 & 2 \\ 0 & 6 & 4 \\ 0 & 0 & 6 \end{bmatrix} \xrightarrow{\mathbf{R}}$$

Theorem. QR Decomposition – Explicit Form.

Let $\mathbf{U}$ be a ~~square~~ matrix whose columns are linearly independent.

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be the column vectors of $\mathbf{U}$.

Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be the vectors produced by Gram-Schmidt.

Set $\mathbf{Q} = \left[\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \dots, \frac{\mathbf{v}_n}{\|\mathbf{v}_n\|} \right]$ and set $\mathbf{R} = \begin{bmatrix} \|\mathbf{v}_1\| & \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|} & \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|} & \dots \\ 0 & \|\mathbf{v}_2\| & \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|} & \dots \\ 0 & 0 & \|\mathbf{v}_3\| & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$.

Then $\mathbf{Q}$ is orthogonal matrix and $\mathbf{R}$ is upper triangular and $\mathbf{U} = \mathbf{QR}$.

Definition.

A matrix $\mathbf{Q}$ is *orthogonal* if its column vectors are orthonormal. Equivalently, we have

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{I}.$$

Not recommended
to memorize

What can be done?

Orthogonal/Orthonormal Basis

- Extract **coefficients** or derive **coordinate vector**
 - Tut 10: Q1, Q3, Q4
- Find **orthogonal projection**
 - Tut 10: Q2, Q9

Gram-Schmidt Procedure

- Find **orthonormal basis**
 - Tut 10: Q3, Q10
- Find **QR decomposition**
 - Tut 10: Q3
- Find **orthogonal complement**
 - Tut 10: Q6

$Q9 + Q10$: Inner product of functions.

Define inner products for arbitrary vectors
→ functions
→ $\mathbb{R}^n$, another inner product

General Inner Products

Definition. Let $\mathcal{V}$ be a real vector space. A *real-valued product function* is a function whose domain is $\mathcal{V} \times \mathcal{V}$ and codomain is $\mathbb{R}$.

$$\text{i.e. } \langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$$

A product function is an inner product if it obeys the properties:

- **(Positivity)** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ for all $\mathbf{v} \in \mathcal{V}$.
- **(Definiteness)** $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
- **(Additivity in first slot)** $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$.
- **(Homogeneity in first slot)** $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{R}$.
- **(Symmetry)** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

Example. Consider the vector space $\mathbb{R}^n$ and recall the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + \cdots + x_n y_n.$$

Show that the Euclidean inner product is an inner product.

Definition. A product function is a **real inner product** if it obeys the properties:

- **(Positivity)**

$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \text{ for all } \mathbf{v} \in \mathcal{V}.$$

- **(Definiteness)**

$$\langle \mathbf{v}, \mathbf{v} \rangle = 0 \text{ if and only if } \mathbf{v} = \mathbf{0}.$$

- **(Additivity in first slot)**

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle.$$

- **(Homogeneity in first slot)**

$$\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle.$$

- **(Symmetry)**

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle.$$

- Last week:
we showed Prop X.

- This week:
Prop X implies that
Euclidean inner product
is an inner product.

Proposition X. The Euclidean inner product obeys the following properties:

- **(Positivity)**

$$\langle \mathbf{x}, \mathbf{x} \rangle \geq 0 \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

- **(Definiteness)**

$$\langle \mathbf{x}, \mathbf{x} \rangle = 0 \text{ if and only if } \mathbf{x} = \mathbf{0}.$$

- **(Additivity in first slot)**

$$\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$$

- **(Homogeneity in first slot)**

$$\langle \lambda \mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle$$

- **(Symmetry)** $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

Definition. Let $\mathcal{V}$ be an inner product space. A *norm* is a function whose domain is $\mathcal{V}$ and codomain is $\mathbb{R}$ such that

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

Two vectors u and v are *orthogonal* if $\langle u, v \rangle = 0$.

Proposition. Consider any norm.

Suppose $v \in \mathcal{V}$.

- (a) $\|v\| = 0$ if and only if $v = \mathbf{0}$.
- (b) $\|\lambda v\| = |\lambda| \|v\|$ for $\lambda \in \mathbb{R}$.

Pythagoras Theorem. If u and v are orthogonal,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Definition. Consider the vector space $\mathbb{R}^n$ and the Euclidean inner product of x and y is given by

$$\langle x, y \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

The Euclidean norm of x is given by $\|x\| \triangleq \sqrt{\langle x, x \rangle}$

Two vectors x and y are *orthogonal* if $\langle x, y \rangle = 0$.

* Proof of Prop + Pythagoras Theorem is IDENTICAL to LAST WEEK

① Go through proofs and check them yourself!

— Tut 10 Q7: ask you to check results you proved in Tut 9.

Example. Let $\mathcal{C}_{\mathbb{R}}[a, b]$ be the vector space of all continuous functions whose domain is the interval $[a, b]$. Consider the following product function given by

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

Then, the product is an inner product $\rightarrow$ Proof on colab $\Rightarrow$ relies on calc results

Set $[a, b] = [-\pi, \pi]$. Compute $\langle 1, x \rangle$ and $\langle 1, x^2 \rangle$.

$$\langle 1, x \rangle = \int_{-\pi}^{\pi} 1 \cdot x dx = \int_{-\pi}^{\pi} x dx = \left[\frac{x^2}{2} \right]_{-\pi}^{\pi} = \frac{\pi^2}{2} - \frac{(-\pi)^2}{2} = 0$$

$$\langle 1, x^2 \rangle = \int_{-\pi}^{\pi} 1 \cdot x^2 dx = \dots = \frac{2\pi^3}{3}$$

Example. Consider $\mathcal{C}_{\mathbb{R}}[-\pi, \pi]$ and the product function given by

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x) dx.$$

What is $\|1\|$ and $\|x\|$?

Definition. Let $\mathcal{V}$ be a vector space which has an inner product.

A *norm* is a function whose domain is $\mathcal{V}$ and codomain is $\mathbb{R}$ such that

$$\|v\| \triangleq \sqrt{\langle v, v \rangle}.$$

$$\|1\|^2 = \langle 1, 1 \rangle = \int_{-\pi}^{\pi} 1 dx = 2\pi$$

$$\therefore \|1\| = \sqrt{2\pi} \approx 2.50 \dots$$

$$\|x\|^2 = \langle x, x \rangle = \int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^3}{3}$$

$$\therefore \|x\| = \sqrt{\frac{2\pi^3}{3}} \approx 4.54 \dots$$

So, $|$ is "closer" to 0 than x .

Example. Consider the product function on $\mathcal{C}_{\mathbb{R}}[-\pi, \pi]$ given by

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x) dx.$$

Consider $f(x) = x^2$ and $g(x) = x^3$. Are $f(x)$ and $g(x)$ orthogonal?

Definition. Let $\mathcal{V}$ be a vector space which has an inner product.

Two vectors u and v are orthogonal if $\langle u, v \rangle = 0$.

$$\langle f(x), g(x) \rangle = \langle 1, x \rangle = 0$$

$\therefore f(x)$ and $g(x)$ are orthogonal.

$$\int_{-\pi}^{\pi} x dx = \left[\frac{x^2}{2} \right]_{-\pi}^{\pi}$$

$$\int_{-\pi}^{\pi} x dx = \left[\frac{x^2}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{\pi^2}{2} - \left(\frac{\pi^2}{2} \right) = 0$$

Pythagoras Theorem. If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

Example. Consider the product function on $C_{\mathbb{R}}[-\pi, \pi]$ given by

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x) dx.$$

$f(x) = x^2$ and $g(x) = x^3$ are orthogonal.

What is $\|f + g\|^2$?

$$\|f+g\|^2 = \langle 1+x, 1+x \rangle = \int_{-\pi}^{\pi} (1+x)^2 dx$$

let's use pythagoras Theorem.

$$\|f\|^2 = \|1\|^2 = \dots = 2\pi$$

$$\|g\|^2 = \|x\|^2 = \dots = \frac{2\pi^3}{3}$$

$$\therefore \|f+g\|^2 = 2\pi + \frac{2\pi^3}{3}$$

Proj formulas
still work
see tutorial 10

“Non-Examples” and Examples of Inner Product...

→ Focus on $\mathbb{R}^3$

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = 3x_1y_1 + 4x_2y_2 - 5x_3y_3.$$

Show that this is NOT an inner product.

$$\langle \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \rangle = 3x_1y_1 + 4x_2y_2 - 5x_3y_3$$

Sum fails here so
not inner product

wooclap

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

$$\langle v, v \rangle \geq 0 \text{ for all } v.$$

- (Definiteness)

$$\langle v, v \rangle = 0 \text{ if and only if } v = 0.$$

- (Additivity in first slot)

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

Choose $\underline{z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\begin{aligned} \langle \underline{z}, \underline{z} \rangle &= \langle \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rangle \\ &= 3 \cdot 0^2 + 4 \cdot 0^2 - 5 \cdot 1^2 \\ &= -5 \end{aligned}$$

Positivity
does not
hold
 $\Rightarrow$ not inner
product

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 + \underline{x_1 y_2}.$$

Show that this is NOT an inner product.

wooclap

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

$$\langle v, v \rangle \geq 0 \text{ for all } v.$$

- (Definiteness)

$$\langle v, v \rangle = 0 \text{ if and only if } v = \mathbf{0}.$$

- (Additivity in first slot)

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

Choose $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rangle = 0 + 0 + 0 + 1 \cdot 1 = 1$$

$$\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rangle = 0 + 0 + 0 + 0 = 0$$

But $1 \neq 0$.

Symmetry fails.

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = 3x_1y_1 + 4x_2y_2 + 5x_3y_3 + 1.$$

Show that this is NOT an inner product.

wooclap

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

$$\langle v, v \rangle \geq 0 \text{ for all } v.$$

- (Definiteness)

$$\langle v, v \rangle = 0 \text{ if and only if } v = 0.$$

- (Additivity in first slot)

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

Definiteness does NOT hold. $x = 0 \not\Rightarrow \langle x, x \rangle = 0$
Choose $\tilde{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\langle \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rangle = 3 \cdot 0 + 4 \cdot 0 + 5 \cdot 0 + 1 = 1 \neq 0$$

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = 3x_1y_1 + 4x_2y_2 .$$

Show that this is NOT an inner product.

wooclap

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

$$\langle v, v \rangle \geq 0 \text{ for all } v.$$

- (Definiteness)

$$\langle v, v \rangle = 0 \text{ if and only if } v = \mathbf{0}.$$

- (Additivity in first slot)

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

Definiteness does NOT hold. $\langle \tilde{x}, \tilde{x} \rangle = 0 \Rightarrow \tilde{x} = \mathbf{0}$

Choose $\tilde{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\langle \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rangle = 3 \cdot 0 + 4 \cdot 0 = 0$$

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 + x_1 y_2 + x_2 y_1.$$

Show that this is NOT an inner product.

wooclap

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0 \text{ for all } \mathbf{v}.$$

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$$\langle \mathbf{v}, \mathbf{v} \rangle = 0 \text{ if and only if } \mathbf{v} = \mathbf{0}.$$

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$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle.$$

- (Symmetry)

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle.$$

Definiteness does NOT hold.

Choose $\mathbf{z} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$$\langle \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \rangle = 1 + 1 + 0 - 1 - 1 = 0$$

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3 + x_1 y_2 + x_2 y_1.$$

Show that this is NOT an inner product.

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

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- (Additivity in first slot)

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- (Homogeneity in first slot)

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

How to come up with this counterexample?

let's try to show $\langle x, x \rangle = 0$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + 2x_1 x_2 = 0$$

$$(x_1 + x_2)^2 + x_3^2 = 0$$

$$\Rightarrow x_1 + x_2 = 0 \text{ and } x_3 = 0$$

$$\nRightarrow x_1 = x_2 = x_3 = 0$$

$$\therefore \text{Choose } x_1 = -x_2$$

So for
prev.
example
we chose
 $x_1 = 1, x_2 = -1,$
 $x_3 = 0$

Example. Consider the vector space $\mathbb{R}^3$ and the following product function

$$\langle x, y \rangle = 2x_1y_1 + x_2y_2 + x_3y_3 + x_1y_2 + x_2y_1.$$

Show that this is an inner product.

Definition. A product function is a **real inner product** if it obeys the properties:

- (Positivity)

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- (Symmetry)

$$\langle u, v \rangle = \langle v, u \rangle.$$

(Definiteness) - when $\underline{x} = \underline{0}$, we have $\langle \underline{x}, \underline{x} \rangle = 0$

— Suppose $\langle \underline{x}, \underline{x} \rangle = 0$

$$\text{i.e. } 2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 = 0$$

$$\Rightarrow x_1^2 + (x_1 + x_2)^2 + x_3^2 = 0$$

$$\Rightarrow x_1 = 0, x_1 + x_2 = 0, x_3 = 0$$

$$\therefore x_1 = x_2 = x_3 = 0$$

$$\therefore \underline{x} = \underline{0}$$

$\therefore$ Definite!

Other
properties
DIY

Example. Let $\mathcal{C}_{\mathbb{R}}[a, b]$ be the vector space of all continuous functions whose domain is the interval $[a, b]$. Consider the following product function given by

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

Show that this is an inner product.

Definition. A product function is a **real inner product** if it obeys the properties:

- **(Positivity)**

$$\langle v, v \rangle \geq 0.$$

- **(Definiteness)**

$$\langle v, v \rangle = 0 \text{ if and only if } v = \mathbf{0}.$$

- **(Additivity in first slot)**

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- **(Homogeneity in first slot)**

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- **(Symmetry)**

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Show that this is an inner product.

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- **(Additivity in first slot)**

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- **(Homogeneity in first slot)**

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- **(Symmetry)**

$$\langle u, v \rangle = \langle v, u \rangle.$$

Example. Let $\mathcal{C}_{\mathbb{R}}[0,1]$ be the vector space of all continuous functions whose domain is the interval $[0,1]$. Consider the following product function given by

$$\langle f, g \rangle = \int_0^{1/2} f(x)g(x) dx.$$

Show that this is NOT an inner product.

Definition. A product function is a **real inner product** if it obeys the properties:

- **(Positivity)**

$$\langle v, v \rangle \geq 0.$$

- **(Definiteness)**

$$\langle v, v \rangle = 0 \text{ if and only if } v = \mathbf{0}.$$

- **(Additivity in first slot)**

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle.$$

- **(Homogeneity in first slot)**

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle.$$

- **(Symmetry)**

$$\langle u, v \rangle = \langle v, u \rangle.$$

Two More Weeks

- **Week 12**

- Lec12A: BTS Tut10
- Lec12B: AY 2021-2022 Paper. Q3. (inner products)

- **Week 13**

- Lec13A: AY 2021-2022 Paper. Q1. (Linear maps)
- Lec13B: AY 2021-2022 Paper. Q2 and Q4. (eigen vectors + eigen values)

Past Year Papers are here:

https://colab.research.google.com/drive/1C_JU8ZDFwZ9M5_gg_hdFy_NA4uLybVktM#scrollTo=xlgKvVcMywla